Department of Information Technology

2025-26

H.T.No: | | | | | | | | | | |

Course Code No: 23IT4T02

VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

II B. Tech II Semester (R23) Regular Examinations, April / May – 2025

OPTIMIZATION TECHNIQUES
(IT, AI & DS, AI & ML)

Time:3hours **Max. Marks: 70M**

Note:1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer all the questions in **Part-A**
3. Answer either a or b questions from Question number 2 to 6 in **Part-B**

PART –A

1	I	What is the difference between constraint surface and a composite constraint surface?	L2	CO1	[2M]
	II	What is the significance of Lagrange multipliers?	L1	CO2	[2M]
	III	What is pivot operation?	L1	CO3	[2M]
	IV	What is basic degenerate solution?	L1	CO4	[2M]
	V	Mention two methods used to find the initial basic feasible solution in a transportation problem.	L1	CO5	[2M]
	VI	Explain the steps involved in converting an unbalanced transportation problem into a balanced one.	L2	CO1	[2M]
	VII	Explain univariate method.	L1	CO2	[2M]
	VIII	Explain exterior penalty function method	L1	CO3	[2M]
	IX	What is multistage decision problem?	L1	CO4	[2M]
	X	How many state variables are to be considered if an LP problem with n variables and m constraints to be solved as a dynamic programming problem?	L1	CO5	[2M]

PART –B

UNIT – I

2(a)	I	Find the maxima and minima, if any, of the function $f(x) = 2X_1^2 + 4X_2^2$ subject to $3x_1 + 2x_2 = 12$, $x_1, x_2 \geq 0$	L1	CO2	[5M]
	II	Explain necessary and sufficient for maximum and minimum of a multivariable function f(X) with equal constraints.	L2	CO2	[5M]
		OR			
2(b)		Consider the following optimization problem: Maximize f = (x+y), subject to $x^2+y \geq 2$, $x+3y \geq 4$ and $x+y^4 \leq 30$ Find whether the design vector X=(1,1) satisfies the Kuhn-Tucker conditions for a constrained optimum.	L3	CO3	[10M]

UNIT – II

3(a)	I	Solve the system of equations using pivot operations: 6x-2y+3z=11	L2	CO5	[6M]

$4x+7y+z=21$
$5x+8y+9z=48$

II Explain : (a) Hyper line (b) Convex set (c) Feasible solution (d) Optimal solution — L1 CO5 [4M]

OR

3(b) *Maximize* $f=x+3y$ — L2 CO1 [10M]
Subject to
$-4x+3y\leq 12$
$x+y\leq 7$
$x-4y\leq 2$
$x, y \geq 0$
Using Simplex method.

UNIT – III

4(a) Solve the transportation problem for the optimal solution. — L2 CO5 [10M]

		A	B	C	D	E	Available
Factory	P	4	1	2	6	9	120
	Q	6	4	3	5	7	120
	R	5	2	6	4	8	120
Demand		40	50	70	90	90	

OR

4(b) *Solve the transportation problem for the optimal solution* — L2 CO1 [10M]

		P	Q	R	Available
Factory	I	5	4	3	250
	II	8	4	3	300
	III	9	7	5	300
Demand		300	200	200	

UNIT – IV

5(a) I Find the minimum of the function $f(x) = x^5-5x^3-20x+5$ by Fibonacci search method in the interval (0, 5) up to 6 iterations. — L1 CO5 [10M]

OR

5(b) I Why the penalty function methods are also known as sequential unconstrained minimization techniques — L3 CO1 [3M]

II Explain the methods of interior and exterior penalty function methods to minimize f. — L1 CO5 [7M]

UNIT – V

6 (a) Min $f = x^2 + y^2 + z^2$ subject to the conditions $x+y+z\geq 0$ and $x,y,z \geq 0$ — L3 CO5 [10M]

OR

6(b) I Why are the components numbered in reverse order in dynamic programming — L3 CO5 [4M]

II Explain briefly computational procedure in dynamic programming. — L1 CO5 [6M]

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